Potential divider design

Values measured on the bench after soldering.

Component	Value	Tolerance
R1	4.7 kΩ	± 5%
R2	10 kΩ	± 1%
C1	100 nF	± 10%

The output follows from the divider rule and the power from Ohm's law.

$$V_{out} = V_{in} \times R_2 / (R_1 + R_2)$$

$$P = I^2 R$$